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8. Simplify $(5x-3y)^2-(5x+3y)^2$

9. Simplify: (i) $(a+b)^2-(a-b)^2$ (ii) $(a+b)^2+(a-b)^2$

10. A square lawn has a 2 m wide path surrounding it. If the area of the path is 136 m^2, find the area of lawn.

11. Solve the following inequalities.

    (i) $4n + 7 \geq 3n + 10$, n is an integer.

    (ii) $6(x + 6) \geq 5(x - 3)$, x is a whole number.

    (iii) $-13 \leq 5x + 2 \leq 32$, x is an integer.

Summary

- The following identities are proved geometrically:
  - $(x + a)(x + b) = x^2 + x(a + b) + ab$
  - $(a+b)^2 = a^2 + 2ab + b^2$
  - $(a-b)^2 = a^2 - 2ab + b^2$ and
  - $(a + b)(a - b) = a^2 - b^2$.
- The factors of an algebraic expression is two or more expressions whose product is the given expression.
- The process of writing an algebraic expression as the product of its factors is called factorisation.
- An algebraic statement that shows two algebraic expression being unequal is known as an algebraic inequation.
- The algebraic expressions are connected with any one of the four signs of inequalities, namely, $>, \geq, <$ and $\leq$.
- When both sides of an inequation are added, subtracted, multiplied and divided by the same non-zero positive number, the inequality remains the same.
- When both sides of an inequation are multiplied or divided by the same non-zero negative number, the sign of inequality is reversed. For example, $x < y \Rightarrow -x > -y$.
- The solutions of an inequation can be represented on the number line by marking the true values of solutions with different colour on the number line.

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